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Education

Indian Institute of Technology, Kharagpur

Aug. 2023 – May 2027

B.TECH in Aerospace Engineering

8.75/10

Relevant Coursework

- Programming and Data Structures
- Linear Algebra
- Advanced Calculus
- Transform Calculus
- AI Foundations and Applications
- AI/ML for Robot Autonomy
- Intro to Flight Vehicle Controls
- Automatic Control of Aircraft
- Dynamics for Aerospace Engineers
- * Aircraft Stability and Control
- Embedded Control and Software
- * MPC with Applications

*mentioned with * are in progress*

Experience

Robotic Research Centre, IIITH

Dec. 2025 – Present

Research Intern

- Built a nonlinear 6-DOF fixed-wing UAV model in Python and validated it to support RL-based control design.
- Designed a Gym based environment and trained a PPO agent to hold/climb to the desired altitude by reward shaping.
- Implemented a safety filter with stall avoidance constraints using control barrier functions within the control framework.
- Developing a model-free multi-agent framework to control the fixed-wing UAV under wind gusts at high altitudes.

CSIR-National Aerospace Laboratories (NAL)

May 2025 – Jul. 2025

Research Intern

- Conducted a literature survey on global high-altitude airship projects to identify suitable modeling assumptions.
- Derived 6-DOF equations of motion to account for added mass and inertia tensors unique to buoyant vehicles.
- Modeled the complete 6-DOF nonlinear flight dynamics model of a high-altitude airship in MATLAB/Simulink.
- Identified trim conditions under suitable assumptions and evaluated open-loop stability of the linearized model.

Projects

Model Predictive Control for Spacecraft Proximity Operations

Feb. 2026 - Present

- Designed a Model Predictive Controller for spacecraft rendezvous using the Hill-Clohessy-Wiltshire equations.
- Enforced safety-critical constraints such as Line-of-Sight approach cone and thruster limits using YALMIP.
- Guaranteed recursive feasibility and stability by synthesizing a Terminal Control Invariant Set via MPT3 toolbox and a terminal cost matrix P derived from the Discrete Algebraic Riccati Equation.

3D Encirclement and Capture with unknown evader strategies | Research project

Nov. 2025 - Present

- Investigated a volume-based approach for encirclement of an evader in a three-dimensional unbounded environment.
- Developed a framework to identify the enclosing tetrahedron of the evader among multiple pursuers using Delaunay triangulation and the visibility walk algorithm.
- Designed a hybrid pursuit strategy combining pure pursuit within the convex hull and Rodrigues-based rotational reorientation (with rotation angle parameterized by evader velocity) upon hull face intersection.

Schlieren Imaging & Supersonic Aerodynamic Analysis | Research Project

Sep. 2025 - Dec. 2025

- Conducted high-speed flow visualization of scaled Concorde and Sukhoi Su-57 nose sections at Mach 2.2 across varying angles of attack (0° to 10°). Prepared CAD models of the nose sections and fabricated test models via CNC machining.
- Compared shock angles from Schlieren images against theoretical predictions from the Taylor-Maccoll theory.
- Analyzed the impact of chined geometry of Su-57 on shock curvature, identifying deviations from the Concorde.

PID Controller for Quadcopter Altitude Control | Self-Project

Nov. 2024

- Modeled vertical dynamics and designed an altitude control system in Simulink to achieve stable altitude hold.
- Simulated real-world conditions by incorporating actuator saturation and sensor noise to assess controller robustness.
- Iteratively tuned PID gains to minimize overshoot, improve settling time, and reduce steady-state tracking error.

Technical Skills

Languages: Python, C, MATLAB, LaTeX

Tools and Libraries: Numpy, Pandas, Matplotlib, PyTorch, Gymnasium, Stable-Baselines3, YALMIP, CVXOPT, Git

Software and Simulation: PyBullet, ROS, Linux, Simulink, SolidWorks, Fusion 360, ANSYS

Leadership / Extracurricular

- Member of the Mechanical Subsystem at TeamTrak. Contributing to the design and development of a miniature locomotive to participate in the IMechE Railway Challenge (UK)
- Iteratively designed and tested three RC aircraft to meet Boeing's mission-specific payload requirements. Ranked in the Top 30 nationally in the Boeing National Aeromodelling Competition.
- Built and programmed an RC car using ESP32 and secured 2nd place in the Formula RC competition at IIT Kharagpur.
- Organised the 7th edition of Local Startups' Meet Chennai 2024, facilitating participation of 30+ startups, 20+ investors, and 5+ VCs.